

### Homework (Carolina Reaper)

- Q1.** The inequality  $2x + 3y \geq 12$  describes the region of the canvas for a painting where the artist will use a specific brush stroke. What type of line best represents the boundary of this region?
- (A) Solid line  
(B) Dashed line  
(C) Dotted line  
(D) Dash-dot line  
(E) None of these
- Q2.** A rhythm pattern is programmed into a music machine using the frequency  $f$  and time  $t$ . The linear inequality  $f + 2t \leq 20$  must be satisfied. What is the  $f$ -intercept of the boundary line?
- (A) 10  
(B) 20  
(C) 0  
(D) -10  
(E) -20
- Q3.** The solution to the linear inequality  $y \geq 3x - 2$  is best described as the region
- (A) Above and including the line  $y = 3x - 2$   
(B) Below and including the line  $y = 3x - 2$   
(C) Above but not including the line  $y = 3x - 2$   
(D) Below but not including the line  $y = 3x - 2$   
(E) None of these
- Q4.** A music teacher wants to spend  $x$  hours practicing violin and  $y$  hours practicing piano, with the constraint that  $x + 2y \leq 6$ . The solid line in the graph best represents the boundary of this region if
- (A) Equality holds  
(B) Inequality holds strictly  
(C)  $x$  and  $y$  are non-negative  
(D)  $x \geq 2y$   
(E)  $y \leq 2x$
- Q5.** If the inequality  $3x - 4y \geq 12$  is graphed on a coordinate plane, the region below and including the line  $3x - 4y = 12$  is shaded. What type of line should be used for the boundary?
- (A) Dashed line  
(B) Solid line  
(C) Dotted line  
(D) Dash-dot line  
(E) None of these
- Q6.** An art piece consists of squares of side length  $x$  and triangles of side length  $y$ , where  $x + y \leq 8$ . If the inequality holds as an equality, the line representing the boundary has the equation
- (A)  $x + y = 8$   
(B)  $x - y = 8$   
(C)  $x + 2y = 8$   
(D)  $2x + y = 8$   
(E)  $y = 8 - x$
- Q7.** An artist creates a geometric pattern using the linear inequality  $y \leq 2x + 1$ . If the area below the line  $y = 2x + 1$  is shaded, what is true about the region?
- (A) The line  $y = 2x + 1$  is included in the region  
(B) The line  $y = 2x + 1$  is excluded from the region  
(C) The line  $y = 2x + 1$  is the region  
(D) The line  $y = 2x + 1$  is parallel to the region  
(E) None of these
- Q8.** The region described by  $2y \geq x + 3$  can be represented on a graph with a
- (A) Dashed line and shaded region above it  
(B) Dashed line and shaded region below it  
(C) Solid line and shaded region above it  
(D) Solid line and shaded region below it  
(E) None of these

#### Open Ended Questions (FRQ)

- Q9.** The linear inequality  $y \geq x - 4$  describes the region where a specific audio frequency  $f$  will not cause sound distortion. Graph the inequality and identify the region where  $f$  will not cause distortion, if  $y$  represents the frequency and  $x$  represents the time in seconds. Provide a brief explanation of your graph.
- Q10.** An artist is designing a canvas of size  $x$  by  $y$ , with the constraint that  $2x + y \leq 18$ . Graph the inequality and find the maximum area  $A = xy$  of the canvas, where the boundary line represents the maximum size allowed. Show all steps to solve for the maximum area.

#### Chef's Tips

- Tip 1: Ensure students understand that the direction of the inequality affects the type of line and the region.
- Tip 2: Encourage students to carefully analyze the inequality to determine the correct type of line (solid or dashed) for the boundary.
- Tip 3: Remind students to always label their axes and provide clear explanations for their answers.